

A linear face bound already follows from three-wise fairness

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Here three-wise permutation fairness means that every restriction to at most three dice is permutation-fair. The dice are independent, uniform on their finite face sets, and all labels across dice are distinct. Write c_i for the number of faces of die i . These conclusions are much weaker than the new per-die lower bound for full permutation fairness; their point is that only triple restrictions are used.

Theorem 1. *Every three-wise permutation-fair collection of $n \geq 3$ dice satisfies*

$$\sum_{i=1}^n c_i \geq \binom{n}{2}.$$

If every die has the same number M of faces, then either $M \geq n - 1$, or $M \geq (n + 3)/2$. In particular, for $n \geq 5$ one has $M \geq (n + 3)/2$.

Let F_j be the cumulative distribution function of die j , and define

$$d_{ij} = \mathbb{E}[F_j(X_i)^2] - \frac{1}{3} \quad (i \neq j).$$

Pair fairness gives $\mathbb{E}F_j(X_i) = 1/2$, so this is also $\mathbb{E}(F_j(X_i) - 1/2)^2 - 1/12$.

Lemma 2 (A positive discrepancy for every pair). *For two pair-fair dice with disjoint labels,*

$$d_{ij} + d_{ji} = \frac{1}{2} \int_{\gamma} (x - y)^2 d(x + y) \geq \frac{1}{24} (c_i^{-2} + c_j^{-2}) > 0,$$

where γ is the axis-parallel path of their two cumulative distribution functions, from $(0, 0)$ to $(1, 1)$.

Proof. At an i -face interpolate the jump of F_i by a horizontal segment; at a j -face use a vertical segment. Distinct labels make these steps unambiguous. Since the other coordinate is constant during each step,

$$\int_{\gamma} y^2 dx = \mathbb{E}F_j(X_i)^2, \quad \int_{\gamma} x^2 dy = \mathbb{E}F_i(X_j)^2.$$

Put $A = \int_{\gamma} (y^2 dx + x^2 dy)$. The identities

$$\int_{\gamma} (x^2 dx + y^2 dy) = \frac{2}{3}, \quad \int_{\gamma} d(x^2 y + x y^2) = 2$$

give

$$\int_{\gamma} (x - y)^2 d(x + y) = 2A - \frac{4}{3}.$$

This proves the equality. On a horizontal segment of length a , the integral of $(x - y)^2$ is at least $a^3/12$, with the minimum attained when the segment is centered at $x = y$. There are c_i such segments of length $1/c_i$. The analogous vertical estimate proves the lower bound. $\square$

Proof of Theorem 1. Fix i . Regard the centered vectors $v_j = (F_j(x) - 1/2)_x$ a face of i , $j \neq i$, as vectors in the mean-zero subspace of $\mathbb{R}^{c_i}$, with normalized inner product. For distinct $j, k \neq i$, three-wise fairness gives

$$\mathbb{E}F_j(X_i)F_k(X_i) = \Pr(X_j < X_i, X_k < X_i) = \frac{1}{3}.$$

Consequently their Gram matrix is

$$G_i = D_i + \frac{1}{12}\mathbf{1}\mathbf{1}^T, \quad D_i = \text{diag}(d_{ij} : j \neq i).$$

Since $\text{rank } G_i \leq c_i - 1$, the diagonal matrix D_i has rank at most c_i . Thus at most c_i entries d_{ij} are nonzero. By Lemma 2, each unordered pair $\{i, j\}$ supplies at least one nonzero directed entry, giving $\binom{n}{2} \leq \sum_i c_i$.

For the refinement, assume $c_i = M < n - 1$ for every i . The rank bound forces a zero entry $d_{ia} = 0$ in each row. Write $\rho = 1/12$ and $e_j = v_j - v_a$. Then $\|v_a\|^2 = \rho$, all e_j are orthogonal to v_a , and distinct e_j are mutually orthogonal, because all off-diagonal Gram entries equal ρ . Moreover $d_{ij} = \|e_j\|^2 \geq 0$. The vectors live in a space of dimension $M - 1$, and $v_a \neq 0$, so at most $M - 2$ of the d_{ij} are positive. Again every unordered pair has a positive directed entry. Hence $\binom{n}{2} \leq n(M - 2)$, as claimed. $\square$

Implication for block constructions. A universal collection of short blocks which is already exactly three-wise fair cannot have a bounded number of faces per die as the alphabet size grows. In particular, replacing the two-face palindromic blocks in the random approximation construction by exact three-wise fair blocks costs at least linearly many faces. This rules out that particular constant-cost upgrade; it is not a lower bound for general approximate fairness.